COMMERCIAL ITEM DESCRIPTION

OSCILLOSCOPE

The General Services Administration has authorized the use of this commercial item description.

Scope. Intended use is for common Aerospace Ground Equipment, Foreign Military Sales.

Applicable documents: MIL-T-28800 - Test Equipment for Use with Electrical and Electronic Equipment, General Specification.

Salient characteristics.

1. Cathode-Ray Tube and Controls

a. Main Frame

(1) Type: post-accelerator storage; approximately 8.5 kV accelerating potential; aluminized P31 phosphor.

(2) Graticule: 8 X 10 div internal graticule, 0.2 subdivision markings on major horizontal and vertical axes 1 div = 0.95 cm. Font panel adjustment aligns trace with graticule.

(3) Beam Finder: returns trace to CRT screen regardless of horizontal or vertical control setting.

(4) Intensity Modulation (external input) Input Approx +2V, $\geq$ 50 ns Pulse width ($\leq$10 MH sine wave) will blank trace of normal intensity.

(5) Input R: approx 5k Ω

(6) Maximum Input: $\pm$20V (dc + peak ac).

(7) Persistence

(8) Normal: natural persistence of P31 phosphor (approx 40 μs).

(9) Variable: from $<$0.2s to $>$1 min.

(10) Storage Writing Speec

(11) Write Mode: $>$ 20 cm/ms

(12) Max Write Mode: $>$5cm/μs.

(13) Brightness: $>$342.6 cd/m^2 (100 foot lambert)

(14) Storage Time: from write mode to store, trace may be stored at reduced intensity for $>$1 hour; to view mode, traces may be viewed at normal

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intensity for >1 minute. From Max Write mode to Store, traces may be stored at reduced intensity for >5 minutes; to View mode, traces may be viewed at normal intensity for >15 seconds.

(15) Erase: manual, pushbutton erasure takes approx 300 ms.

b. Horizontal Amplifier

(1) External input

(2) Bandwidth: dc-coupled, dc to 5 MH_z, ac-coupled, 5 H_z to 5 MH_z

(3) Deflection Factor: 1 V/div in X1; 0.2 V/div in X5, 0.1 V/div in X10.

(4) Vernier: provides continuous adjustment between ranges.

(5) Dynamic Range: ±20V.

(6) Maximum input: 600 Vdc (ac-coupled input)

(7) Input RC: approx 1 megohm shunted by approx 30 pF.

(8) Sweep Magnifier: X5, X10; accuracy, ±5% with 3% accuracy time base.

c. Calibrator

(1) Type: approx 1 kHz square wave, 3 μs rise time.

(2) Voltage. 10V p-p into ≥1 megohm, accuracy, ±1%.

d. Outputs

Four rear panel emitter follower outputs for main and delayed gates, main and delayed sweeps or vertical and horizontal outputs when used with TDR/sampling plug-ins. Will drive impedances ≥1000 ohms without distortion.

e. General

(1) Operating Environment

(a) Temperature 0 to +55°C

(b) Humidity: to 95% relative humidity at 40°C.

(c) Altitude: to 4600 m (15 000 ft).

(d) Vibration: vibrated in three planes for 15 min. each with 0.254 mm (0.010 in) excursion, 10 to 55 hz.

(2) Dimensions: for 19" rack mount.

2. Modes of Operation (Plug-in)

a. Channel A; channel B, channels A and B displayed alternately on successive

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sweeps (ALT); channel A and B displayed by switching between channels at approx 500 kHz rate (CHOP), with blanking during switching; channel A plus channel B (algebraic addition).

b. Each Channel (2)

(1) Bandwidth: (Measured with or without 10014A probe, 3dB down from 8 div reference signal from a terminated 50 ohm source.)

(2) DC-Coupled: dc to 100 MHz

(3) AC-Coupled: approx 10 Hz to 100 MHz (lower limit is approx 1 Hz with 10014A probe).

(4) Rise Time: <3.5 ns (measured with or without 10014A probes, 10% to 90% points of 6 div input step from a terminated 50 ohm source)

(5) Deflection Factor

(a) Ranges: 5 mV/div to 5 V/div (10 calibrated positions) in 1, 2, 5 sequence.

(b) Attenuator Accuracy: ±2%

(c) Vernier: provides continuous adjustment between deflection factor settings and extends maximum deflection factor to at least 12.5 V/div. Front panel light indicates when vernier is not in CAL position.

(6) Polarity: + up or -up, selectable.

(7) Signal Delay: input signals are delayed sufficiently to view leading edge of input pulse without advanced trigger.

(8) Input Coupling: AC, DC, 50 ohms (dc), or ground. Ground position disconnects input connector and grounds amplifier input.

(9) Input RC

(10) AC and DC: 1 megohm ±1% shunted by approx 13 pF. Constant on all ranges.

(11) 50 Ohm: 50 ohm ±2% SWR <1.2 at 100 MHz or all ranges.

(12) Maximum Input

(13) AC and DC: ±300V (dc + peak ac) at 1 khz or less. ±150V (dc + peak ac) on 5mV/div range at 1 kHz or less.

(14) 50 ohm: 10V rms.

(15) Dynamic Range: 6 div at 100 MHz increasing to 16 div at ≤ 15 MHz.

(16) Positioning Range: 16 div.

(17) A + B Operation

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(18) Amplifier: bandwidth and deflection factors are unchanged; either channel may be inverted for ±A ±B operation.

(19) Differential Input (A - B) Common Mode: CMRR is at least 40 dB from dc to 1 MHz for common mode signals of 16 div or less. CMRR is at least 20 dB at 50 MHz for common mode signals of 6 div or less.

c. Triggering

(1) Source: selectable from channel A, channel B, or a composite (Comp) signal from A and B in any display mode. Composite is channels A and B signals switched for Alt and Chop modes and added for A and B mode. Vernier and position controls do not affect A, B, or composite trigger signals. A and B signals are independent of polarity selection.

(2) Frequency

Time Base Plug-In	Trigger Frequency*	Required Vertical Deflection
1820C, 1824A 1825A, 1840A, 1841A	dc - 50 MHz	1/2 div
	dc - 100 MHz	1 div
1820B, 1822A	dc - 50 MHz	1/2 div
	dc - 100 MHz	2 div
1820A, 1821A	dc - 50 MHz	1 div

*All display modes except Chop. dc to 100 kHz in Chop.

d. Offset

±200 div of offset. Allows offset of dc or ac signals up to the dynamic range and maximum input.

e. Vertical Signal Output

(selected by trigger source switch)

(1) Bandwidth: >50 MHz into 50 ohms

(2) Amplitude: >50 mV for each division of display into 50 ohms with useable amplitudes up to 500 mV p-p

(3) Source Impedance approx 50 ohms

f. General

(1) Operating Environment

Temperature: 0 to +55°C.

(a) Humidity: to 95% relative humidity at 40°C.

(b) Altitude: to 4600 m (15 000 ft).

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(c) Vibration: vibrated in three planes for 15 min. each with 0.254 mm (0.010 in.) excursion, 10 to 55 Hz.

3. Main Time Base (Plug-in)

a. Sweep

(1) Ranges: from 0.1 μs/div to 1 s/div (22 positions) in 1, 2, 5 sequence. ±3% accuracy with vernier in calibrated position.

(2) Vernier: continuously variable between all ranges; extends slowest sweep to at least 2.5 s/div.

(3) Magnifier: (on mainframe) expands fastest sweep to 10 ns/div.

(4) Normal. sweep is triggered by an internal, external, or power line signal.

(5) Automatic bright baseline displayed in absence of input signal. Triggering same as normal except low frequency limit is 40 Hz for internal or external modes.

(6) Single: sweep occurs once with same triggering as normal, reset pushbutton with indicator light.

b Delayed Time Base. Delayed time base sweeps after a time delay set by Main time base and Delay controls.

(1) Sweep

(2) Ranges. from 0.1 μs/div to 50 ms/div (18 positions) in 1, 2, 5 sequence. ±3% accuracy with Vernier in calibrated position.

(3) Vernier: continuously variable between all ranges, extends slowest sweep to at least 125 ms/div.

(4) Magnifier: (on mainframe) expands fastest sweep to 10 ns/div.

c. Triggering

(1) Main Time Base

(a) Internal: refer to vertical amplifier plug-in specifications.

(b) External: from dc to 50 MHz on signals 0.5V p-p or more, increasing to 100 MHz on signals 1V p-p or more.

(c) Line: power line frequency signal.

(d) Level

INTERNAL: at any point on the vertical waveform displayed.

EXTERNAL: Continuously variable from +3V to -3V on either slope of the sync signal; from +30V to -30V in ÷ 10 setting.

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(e) Slope: switch selection of either positive or negative slope of trigger signal

(f) Coupling: front panel selection of AC, DC, ACF, or ACS.

AC: attenuates signals below approx 20 Hz.

ACF (ac-fast): attenuates signals below approx 15 kHz.

ACS (ac-slow): attenuates signals above approx 30 kHz.

b. Trace Intensification: used when setting up Delayed or Mixed time base. Intensifies that part of Main time base to be expanded to full screen on Delayed time base. Rotating Delayed time base sweep switch from Off position activates intensified mode. Front panel screwdriver adjust sets relative intensity of brightened segment.

c. Delayed Time Base: applies to intensified Main, Delayed, and Mixed time base triggering.

(1) Internal: refer to vertical amplifier plug-in specifications.

(2) Automatic: delayed sweep is automatically triggered at end of set time delay.

(3) External: from dc to 50 MHz on signals 0.5V p-p or more, increasing to 100 MHz on signals 1V p-p or more.

(4) Level

INTERNAL: at any point on the vertical waveform displayed.

EXTERNAL: continuously variable from +3V to -3V on either slope of the sync signal; from +30V to -30V in ÷ 10 setting.

(5) Slope: switch selection of either positive or negative slope of trigger signal.

(6) Coupling: Front panel selection of AC, DC, ACF, or ACS.

AC: attenuates signals below approx 20 Hz.

ACF (ac-fast): attenuates signals below approx 15 kHz.

ACS (ac-slow) Attenuates signals above approx 30 kHz.

(7) Delay (before start of Delayed sweep)

(8) Time: continuously variable from 0.1 μs to 10 s.

(9) Accuracy: $\pm$1%. Linearity, $\pm$0.2%. Time jitter is $<$0.005% (1 part in 20,000) of maximum delay of each step.

(10) Trigger Output: (at end of Delay time) approx 1.5V with $<$50 ns rise time from 1000 ohm source resistance.

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d. Mixed Time Base

Dual time base in which main time base drives first portion of sweep and delayed time base completes sweep at up to 1000 times faster. Also operates in single sweep mode.

e. General

(1) Operating Environment

(a) Temperature: 0 to +55°C.

(b) Humidity. to 95% relative humidity at 40°C.

(c) Altitude: to 4600 m (15 000 ft).

(d) Vibration: vibrated in three planes for 15 min. each with 0.254 mm (0.010 in.) excursion, 10 to 55 Hz.

4. Power 115 or 230V ±10%, 48-440 Hz.

Design: MIL-T-28800, type III, class 5, style E, color R.

Construction: System safety per MIL-T-28800, paragraph 3.2.

Contractor certification. Bid sample shall be submitted to SA-ALC/MMETT (Caston Tesson) Kelly AFB. TX 78241 for acceptance testing.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes. No longer made acceptable item. Hewlett-Packard P/N 1553C (consists of P/N: 181 AR, 1805, 1821A), 6625 01 038 3420.

UL Standard 1236 is available from Underwriters' Laboratories (UL), Inc., 1285 Walt Whitman Road, Melville, Long Island, New York 11747.

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